

Amit Kumar Mahato

Full Stack Developer | Backend Engineering | Scalable Systems
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SUMMARY

- Computer science student and self-taught full stack developer who builds things that run in production.
- Shipped a published browser extension (Pixelva) available on the Microsoft Edge Add-ons Store.
- Architected a multi-tenant SaaS platform (ManageInSync) from scratch — zero to production. Stress-tested Node.js backends to **14,000 req/sec**, documenting every bottleneck and trade-off.
- Core interests: backend engineering, scalable distributed systems, and high-performance web applications.

TECHNICAL SKILLS

Languages:	JavaScript, TypeScript
Frontend:	React.js, Redux, Tailwind CSS, HTML5, CSS3
Backend:	Node.js, Express.js, REST API design, JWT authentication, session management, rate limiting, CORS, CSP hardening
Databases:	MongoDB, Mongoose, Redis (caching + rate limiting)
DevOps & Tools:	Docker, Docker Compose, GitHub Actions CI/CD, NGINX, Render, Vercel
Concepts:	Multi-tenancy, RBAC, distributed systems, load testing, clustering, audit logging

PROJECTS

ManageInSync — Multi-Tenant SaaS Platform | [Live Demo](#) | [GitHub](#)

- Architecture:** Designed the full data model and API surface for an agency SaaS product — clients, projects, teams, invoices, and tasks with hard data isolation between every tenant.
- Auth & RBAC:** Built a four-role permission system (Owner / Admin / Member / Client) enforced at route level, with rotating refresh tokens, HTTP-only secure cookies, and server-side session tracking.
- Billing:** Integrated Razorpay for invoice payments, including webhook verification and idempotent payment-status updates.
- Security:** Hardened the API with per-route rate limits, CORS allowlists, Content-Security-Policy headers, request schema validation, and structured audit logs.
- DevOps:** Containerised with Docker Compose; CI/CD via GitHub Actions deploys to Render on every push to main.
- Performance:** Maintained p99 latency under 60 ms under load using compound indexes, cursor-based pagination, and lean Mongoose projections.

Backend Performance Lab — API Scaling & Benchmarking | [GitHub](#)

- Baseline to 5k:** Started at ~9 req/sec on a naive Express handler; eliminated synchronous I/O, unoptimised middleware, and missing connection pooling to reach ~5,000 req/sec.
- Caching layer:** Added Redis caching for hot routes and sliding-window rate limiting; measured the overhead Redis introduced and documented the trade-off.
- Clustering:** Enabled Node.js cluster mode to saturate all CPU cores, pushing throughput to ~9,700 req/sec and then ~14,000 req/sec at optimal concurrency.
- Methodology:** Every experiment is documented with Autocannon benchmarking output and a clear explanation of why each optimisation improved performance.

Pixelva — Browser Extension (Screenshots & Asset Extraction) | [Edge Store](#) | [GitHub](#)

- Frontend experience:** Built an interactive canvas-based editing interface with draggable elements, text overlays, shape tools, keyboard shortcuts, and real-time visual feedback.
- Asset extraction:** Developed browser-side workflows to collect images, videos, SVGs, PDFs, favicons, and embedded assets via DOM traversal, extension APIs, and resource parsing.

- ⌘ **Extension architecture:** Built the communication layer between content scripts, background service workers, and the editor panel using vanilla JS and a custom event bus.
- ⌘ **Rendering performance:** Maintained 60+ FPS during canvas interaction by batching redraw operations and invalidating only dirty rendering regions.

ACADEMIC QUALIFICATIONS

Degree / Certification	Institute / School	Year
Bachelor of Computer Applications (BCA)	Sarala Birla University, Ranchi	2023 - 2026

CERTIFICATIONS

Job-Ready AI-Powered Bootcamp — Sheryians Coding School

Software Engineering Job Simulation — JPMorgan Chase & Co. (Forage)